Updated: UCL Challenge Outline

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Task outline – trees, clouds and data security

The Carpentries Incubator lesson Introduction to Tree Models in Python steps though an interactive approach to a machine learning model for patient morbidity using the Jupyter Lab.

When developing machine learning models, it is not always possible to work locally in this interactive way, especially when data sets are large. Your task is to develop a script that can be run on the UCL cloud platform Condenser.

For the challenge we will be exploring anti-microbial resistance to antibiotics for E.Coli. The data set, dig_health_hub_amr.csv , is synthetic and is modelled on data used by the UK Health Security Agency.

The columns of the data set of interest are coamox, age, and imd:

- coamox: Values 0 and 1, 1 indicates resistance to coamoxicillin
- age: Calculated age in years at time of testing
- imd: Index of multiple deprivation, quintiles. 1 are the most deprived areas, 5 are the least deprived areas.

You will explore different tree model types and depths.

The script will need to import the data set and save the tree diagrams it will generate in an output folder alongside key model comparison data.

Which tree best identifies those at risk of antibiotic resistance?
What are the main risk factors?

The Carpentries lesson uses IPython.display to visualise the tree diagrams generated via glowyr. You can also plot the surface of a decision tree using sklearn: Plot the decision surface of decision trees trained on the iris dataset

You may prototype locally before up loading and running your script on Condenser.

As part of the marking process, we will be reviewing the contents of your VM alongside a 5-minute video.

Video requirements

In the 5-minute video we would like you to address the following questions:

1. What is cloud compute?
2. What are key data security considerations for cloud computing?
3. The file permissions for the data file enabled global access. Why is this not desirable? Did you change the permissions? If so, how and why?
4. Why would you want a representative data set for machine learning?
5. Why would we want to use synthetic data instead of actual patient data?
6. What is your final tree model, and why do you think this is the most optimal model?
7. What other modelling techniques could you use and why?
8. How did you work as a team?

Submission checklist

- Files on Condenser:
    - Read me

- Python Script
- Data file
- Generated images stored in Outputs folder
- 5-minute video saved as MP4, saved in your team's CIUK Challenge folder

Platform and packages

Each team will have access to an Alma Linux virtual machine on our cloud platform Condenser. You will be sent login details on Monday 26th October, the UCL challenge will run until 23:59 on Friday 30th October. You may complete the challenge at any point during the week.

On Monday 26th October, team captains with received information relating to their team's server. Further details are available in the access instructions.

You will need to create a Python virtual environment with the necessary Python packages to run your script. Reproducibility is a key part of research. As part of your submission, you are required to produce a ReadMe file.

Your ReadMe file will outline:

- How you created your environment
- Access instructions to your environment
- Run instructions for your script

Things to know about the VMs:

- You will not have root permissions
    - For example, this means that you will not be able to install graphviz
- The only available text editor is vim, we recommend adding micro via curl
    - https://micro-editor.github.io/
- Files can be added to your virtual machine using curl.
    - To import the data file please use the following command:
        - curl -L -o dig_health_hub_amr.csv "https://drive.google.com/uc?export=download&id=1KeMv2GaR824SNW2_KDDfCrcSpIObjui0"

Challenge Scoring

This challenge will be scored out of 100.

Scores will be distributed as below:

Scoring Item	Max Points	Quality metric
Technical Solution	**50**	
Accessed Condenser	5	Successfully accessed VM - this will also act as a weighting for the technical solution
Python environment	5	Was a Python 3 environment created, are only required packages installed
ReadMe	10	Clear overview of setup and easy to follow instructions
Python script	20	Is the Python script well structured? Are there useful comments
File permissions and storage	10	Are outputs stored in the requested location? Are file names meaningful?
Video	**35**	
Questions on cloud compute and security	10	Demonstrate an understanding of cloud basics and key security considerations
Questions on data and machine learning	10	Demonstrates an awareness of data protection issues and algorithmic bias
Technical solution explained and alternative analysis methods considered	10	Who is more at risk from AMR? What tree do you think is best and why? How else could you have modelled this data?
Video quality	5	Video is 5 mins and saved as MP4. Audio is clear and any screens shown are easily visible.
Team work	**15**	Explained how the team worked together and discuss any challenges overcome. Evidenced via the video.